



**ITM SKILLS
UNIVERSITY**

CopyrightFlow

Project Report

Student Details

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College:-ITM Skills University, Kharghar

Batch:-2025-2029

Course:BTech (CSE)

Subject:-React JS

Github Repo:- <https://github.com/pratikswain070-blip/CopyrightFlow>

Live Demo Link:- <https://copyright-flow.vercel.app/>

Problem Statement (76)

Problem statement: Create a system for managing the ownership and trading of digital items (like game assets or music licenses), making sure the right people get paid Royalties.

Must Have Features

- a. **1. Digital Asset List:** A database that lists all media tokens and their license rules in a searchable table.
- b. **2. Ownership Undo:** A control that tracks changes in who owns a digital item, allowing safe rollbacks if an ownership transfer fails.
- c. **3. Auction Purchase Queue:** A system that lines up all token transfers and sales from the auction house in the correct time order.
- d. **4. License Checker:** A fast tool that instantly verifies a user's wallet ID against the list of approved ownership certificates.
- e. **5. Trade Volume Sorter:** A tool that ranks different asset trade categories based on how often they are being cleared or transferred.
- f. **6. Global Royalty Hub:** A central system that broadcasts the rules for distributing licensed media across different regions.
- g. **7. Price Safety Finder:** A tool that plans trade validation steps to ensure all payment calculations are correct.
- h. **8. Asset Storage Balancer:** A smart system that divides up digital item descriptions to efficiently save space on the database.

Introduction

The rapid growth of digital content creation has increased the demand for platforms that can efficiently manage ownership rights, licensing agreements, and royalty distribution. Traditional systems often lack transparency and effective tracking mechanisms for digital assets.

CopyrightFlow addresses these challenges by providing a centralized platform where creators can list assets, manage royalties, conduct auctions, and monitor transactions. The platform combines marketplace functionality with royalty management and analytics to deliver a comprehensive digital asset management solution.

Objectives

The primary objectives of the project are:

- 1. Provide a marketplace for digital asset trading.
 - 2. Enable creators to manage assets efficiently.
 - 3. Support royalty-based revenue generation.
 - 4. Facilitate auction-based asset sales.
 - 5. Provide analytical insights through dashboards.
 - 6. Simulate wallet-based ownership management.
 - 7. Deliver a responsive and user-friendly interface.
-

Technology Stack

Frontend Technologies

Technology	Purpose
React 19	User Interface Development
Vite	Build Tool and Development Server
React Router DOM	Client-Side Routing
Tailwind CSS	Styling Framework
Lucide React	Icons
Recharts	Data Visualization
ESLint	Code Quality

System Architecture

The application follows a component-based architecture.

Architecture Flow

User



React Frontend



├── Pages

├── Components

├── Context API

├── Hooks

└── Local Storage



Marketplace Data Layer



Dashboard & Analytics

Project Structure

src/



├── components/

| ├── layout/

| ├── ui/

| └── wallet/



├── pages/



├── context/



├── hooks/



```
|— data/
|
|— App.jsx
|
|— main.jsx
```

Modules Description

1 Marketplace Module

Allows users to:

- Browse digital assets.
- Search and filter listings.
- View asset details.
- Purchase assets.

Features

- Asset listing
 - Asset discovery
 - Ownership information
 - Purchase workflow
-

2 Creator Portal Module

Allows creators to:

- Upload assets.
- Configure royalties.
- Manage ownership.
- Track performance.

Features

- Asset management
 - Royalty configuration
 - Revenue tracking
-

3 Auction Module

Provides auction functionality.

Features

- Auction creation

- Bid management
 - Countdown timers
 - Settlement process
-

4 Dashboard Module

Provides analytical insights.

Features

- Revenue charts
 - Asset performance metrics
 - Transaction summaries
 - Activity tracking
-

5 Wallet Module

Simulates Web3 wallet functionality.

Features

- Wallet connection
 - Session persistence
 - User identification
-

Key Features

Creator Features

- Asset publishing
- Ownership management
- Royalty configuration

Marketplace Features

- Asset browsing
- Search and filtering
- Asset purchasing

Analytics Features

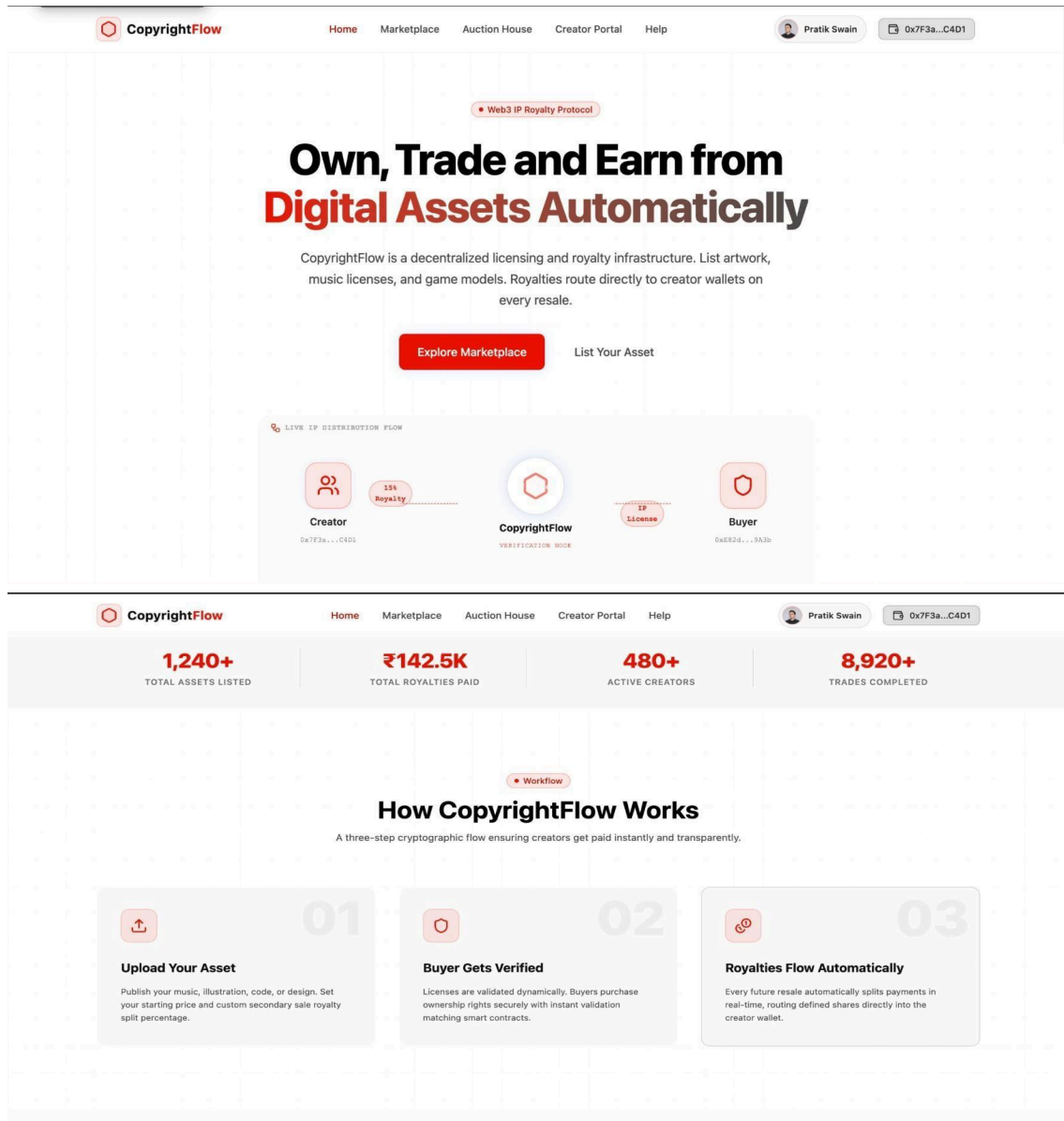
- Earnings overview
- Performance charts
- Activity monitoring

User Experience Features

- Responsive design
- Smooth animations

- Modern UI
- Interactive dashboards

Screenshots



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Capabilities

Engineered for Web3 Fintech

Complete support for automated licensing parameters, escrow splits, and instant safety clearing.

Automated Licensing

Instant minting of intellectual properties as cryptographically secure files.

Instant Splits

Secondary payments route directly to creators without platform delays.

Safety Checker

Real-time double-verification of ownership, prevents copyright fraud.

Multi-Format Support

Easily register music loops, gaming sprite-sheets, typography, and artworks.

Rollback Recovery

Built-in safety protocol reverses assets if transaction failures occur.

Universal Wallets

Compatible with MetaMask, Coinbase Wallet, and WalletConnect.

Royalty Analytics

Track sales, volume, pricing trends, and payouts with clear dashboard reports.

Custom Splits

Define precise royalty splits among co-producers, platforms, and buyers.

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Target Audience

Designed for the Creative Economy

Whether you list original art or build marketplaces, we coordinate royalty transfers.

For Creators

Independent musicians, game developers, designers, and visual artists looking to secure secondary sale earnings.

✓

Guaranteed royalties up to 30%

✓

No middleman fee distributions

✓

Direct connection to buyers

For Buyers & Traders

Producers, project leads, and collectors looking for verified digital materials to embed in their products.

✓

Fully audited, verifiable licenses

✓

Direct transfers in seconds

✓

Secondary market trading

For Enterprise Platforms

Asset libraries, game distributors, and music platforms looking to integrate royalty distribution APIs.

✓

Robust developer webhooks

✓

SLA-backed transfer support

✓

Custom licensing agreements

CopyrightFlow

Automated digital asset licensing and dynamic royalty distribution platform. Secure your intellectual property and earn seamlessly.

PRODUCT

Marketplace

Auction House

Creator Portal

COMPANY

Help Center

Documentation

Terms of Service

NEWSLETTER

Stay updated with the latest licensing updates, auctions, and feature releases.

Your email address

Join

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Asset Marketplace

Explore verified licensing agreements and secondary resell rights.

Filters

CATEGORY

- Game Asset
- Music License
- Artwork
- Sound Effect
- Font License

MAX PRICE (INR)

₹0₹1123

LICENSE TYPE

- Commercial
- Non-Exclusive
- Exclusive
- Personal Use

STATUS

- Available
- Auction
- Sold

Apply Filters

Showing 8 results

Sort by:Trade Volume

Font License

Retro UI Kit v2

TypeHouse

Commercial

Vol: 72%

LICENSE PRICE₹45CRYPTO EQUIVALENT0.017 ETH

View DetailsBuy Now

Music License

Ambient Drift Vol 3

SoundForge

Non-Exclusive

Vol: 62%

LICENSE PRICE₹85CRYPTO EQUIVALENT0.032 ETH

View DetailsBuy Now

Artwork

Urban Skyline 4K Pack

VisualLab

Commercial

Vol: 44%

LICENSE PRICE₹600CRYPTO EQUIVALENT0.231 ETH

View DetailsBuy Now

Sound Effect

Game Asset

Font License

CopyrightFlow

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Auction House

Bid on exclusive licensing agreements and acquire full digital rights.

Live Auctions

Upcoming Auctions

GAME ASSET

03:01:54

Pixel Warrior Sprite Pack

Listed by PixelCo

HIGHEST BID

₹240

BIDS COUNT

5 bids

RECENT BIDS

0x7F3a...C4D1₹240

0xE82d...9A3b₹225

0x3A2b...7C4d₹210

Place Bid

Stop

GAME ASSET

01:30:39

Neon City Background Loop

Listed by Alex Dev

HIGHEST BID

₹120

BIDS COUNT

4 bids

RECENT BIDS

0x3A2b...7C4d₹120

0x7F3a...C4D1₹110

0x5d4A...82e3₹999

Place Bid

Stop

Auction Queue

PIXEL

10x7F3a...C4D12100₹240

Leading

20xE82d...9A3b20:49₹225

Outbid

30x3A2b...7C4d20:30₹210

Outbid

40x987b...32A120:19₹250

FailedRollback

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Automated digital asset licensing and marketplace

PRODUCT

Marketplace

COMPANY

Help Center

NEWSLETTER

Stay updated with the latest licensing

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Creator Portal

Register new digital licenses and configure secondary royalty distributions.

1. ASSET REGISTRY

ASSET NAME *

Neon Horizon Loops

UPLOAD FILE

Drag and drop your file here

Supports ZIP, MP3, WAV, PNG, OTF (Max 50MB)

CATEGORY

Game Asset

DESCRIPTION

A modular, high-tempo synthwave soundtrack pack designed for fast-paced cyberpunk racers.

2. LICENSE FRAMEWORK

Commercial

Exclusive

LIVE CARD PREVIEW

PREVIEW MODE

GAME ASSET

Neon Horizon Loops

My Connected Wallet

Commercial

Vol: 0%

STARTING PRICE

₹120

ROYALTY SPLIT

15% Secondary

Details

Buy Now

Pre-computation fees apply upon listing confirmation. Submitting transfers requires standard wallet approval confirmations.

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2. LICENSE FRAMEWORK

Commercial

Exclusive

Non-Exclusive

Personal Use

LICENSE RULES

Permits embedding in commercial gaming titles and videos. Re-distribution of raw audio files is forbidden.

3. DISTRIBUTION & PRICING

SECONDARY ROYALTY PERCENTAGE

15% Creator Share

0% (Standard)15%30% (Max)

STARTING PRICE (INR) *

₹ 120

LISTING TYPE

Fixed Price

Auction

List Asset on CopyrightFlow

LIVE CARD PREVIEW

PREVIEW MODE

GAME ASSET

Neon Horizon Loops

My Connected Wallet

Commercial

Vol: 0%

STARTING PRICE

₹120

ROYALTY SPLIT

15% Secondary

Details

Buy Now

Pre-computation fees apply upon listing confirmation. Submitting transfers requires standard wallet approval confirmations.

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Documentation Center

How can we help?

Search for answers regarding wallet verification, smart licensing rules, and automatic payout splits.

Q Search help topics, FAQ keywords...

Getting Started Guides

Connecting Your Wallet

Step-by-step setup guides for MetaMask, Coinbase, and WalletConnect extensions.

Buying Your First Asset

Understanding licensing terms, checkout flows, and key registrations.

Listing an Asset

Upload file structures, descriptions, and custom royalty split configurations.

Understanding Royalties

Deep dive into automatic secondary payouts and smart agreements.

Frequently Asked Questions

BUYING AND SELLING

How are licenses delivered to my account?

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MANAGEMENT

Overview

My Assets6

Ownership History10

My Licenses6

Royalty Earnings

Pending Transfers9

Welcome back, Pratik Swain

0x7F3a56241b1285c4d19485c92b950000000c4d1Verified Creator

Request Payout

ASSETS OWNED

6

+8.4% vs last month

ROYALTIES EARNED

₹1161

+12.5% vs last month

PENDING TRANSFERS

0

ACTIVE BIDS

1

ROYALTY EARNINGS (6 MONTHS)

INR ₹

1400

1050

700

350

0

Jan

Feb

Mar

Apr

May

Jun

RECENT ACTIVITY

View Ledger

Resale profit for 'Pixel Warrior Sprite Pack'

Just now

+₹108

Resale profit for 'Urban Skyline 4K Pack'

Just now

+₹100

Resale profit for 'ajit's asset'

Just now

+₹70

Resale profit for 'Neon City Background Loop'

Just now

+₹883

Royalty payout of \$150 processed successfully

Just now

-₹150

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My Licensed Assets

Manage listed inventory and transfer keys

ASSET NAME	CATEGORY	ACQUIRED	PURCHASE PRICE	CURRENT VALUE	LICENSE	STATUS	ACTIONS
project GK4391	Game Asset	2026-06-13	₹300	₹300	Commercial	Active	View List
dsa2 PD3121	Font License	2026-06-13	₹220	₹220	Non-Exclusive	Active	View
dsa AR5392	Artwork	2026-06-13	₹120	₹120	Personal Use	Active	View List
Ambient Drift Vol 3 MC5887	Music License	2026-06-13	₹85	₹85	Non-Exclusive	Active	View
Retro UI Kit v2 PD0963	Font License	2026-06-13	₹45	₹45	Commercial	Active	View
Deep Bass Loops Bundle DQ4793	Sound Effect	2026-06-13	₹60	₹60	Personal Use	Active	View

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Ownership Transfer Log

Cryptographic history records

DATE	ASSET	FROM WALLET	TO WALLET	PRICE	STATUS
2026-06-13	project	0x0000...0000	0x7F3a56241b1285C4D19485C92b9500000000C4D1	₹300	Settled
2026-06-13	dsa2	0x0000...0000	0x7F3a56241b1285C4D19485C92b9500000000C4D1	₹220	Settled
2026-06-13	dsa	0x0000...0000	0x7F3a56241b1285C4D19485C92b9500000000C4D1	₹120	Settled
2026-06-13	Ambient Drift Vol 3	0x3A2b...7C4d	0x7F3a56241b1285C4D19485C92b9500000000C4D1	₹85	Settled
2026-06-13	Retro UI Kit v2	0x2E4b...9C1d	0x7F3a56241b1285C4D19485C92b9500000000C4D1	₹45	Settled
2026-06-13	Deep Bass Loops Bundle	0x9F4c...1B8a	0x7F3a56241b1285C4D19485C92b9500000000C4D1	₹60	Settled
2026-06-05	Retro UI Kit v2	0x2E4b...9C1d	0x7F3a...C4D1	₹45	Settled
2026-06-03	Deep Bass Loops Bundle	0x9F4c...1B8a	0x7F3a...C4D1	₹60	Settled
2026-06-01	Ambient Drift Vol 3	0x3A2b...7C4d	0x7F3a...C4D1	₹85	Settled
2026-05-25	Ambient Drift Vol 3	0x12c4...7D8e	0x3A2b...7C4d	₹75	Settled

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Cryptographic Licensing Keys

Licenses registered with verification signatures

GA4397

project

LICENSE SCOPECommercial

VALID UNTILPermanent

Signature pending

Verify Now

FD3121

dsa2

LICENSE SCOPENon-Exclusive

VALID UNTILPermanent

Signature pending

Verify Now

AB5392

dsa

LICENSE SCOPEPersonal Use

VALID UNTILPermanent

Signature pending

Verify Now

MD5887

Ambient Drift Vol 3

LICENSE SCOPENon-Exclusive

VALID UNTILPermanent

Signature verified

FD5565

Retro UI Kit v2

LICENSE SCOPECommercial

VALID UNTILPermanent

Signature verified

SD4595

Deep Bass Loops Bundle

LICENSE SCOPEPersonal Use

VALID UNTILPermanent

Signature verified

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Secondary Royalty Ledger

Verify distributions split automatically on secondary sales

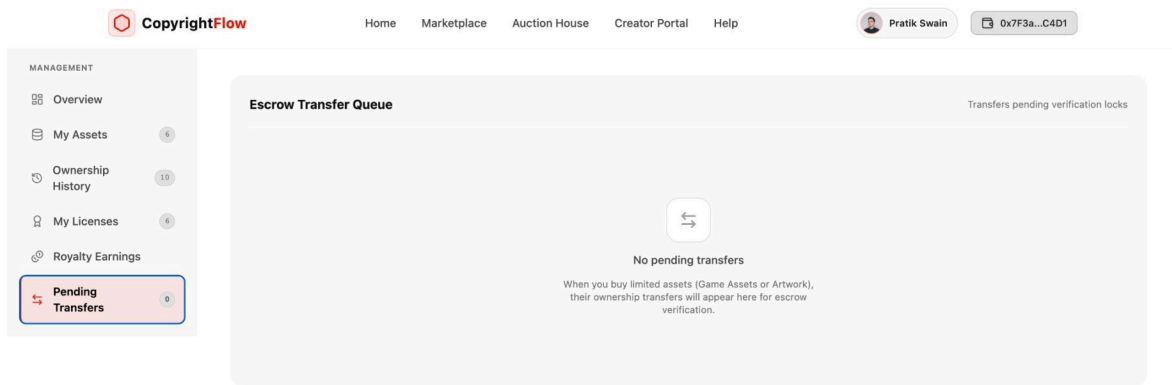
Request Payout

MONTHLY DISTRIBUTIONS



EARNINGS SPLITS BREAKDOWN

ASSET NAME	REGION	DATE	ROYALTY
Neon City Background Loop	US-East	2026-06-09	+₹18.00
Ambient Drift Vol 3	EU-Central	2026-06-08	+₹10.20
Retro UI Kit v2	APAC-East	2026-06-05	+₹4.50



Advantages of the System

- Easy asset management.
- Improved ownership transparency.
- Better royalty tracking.
- User-friendly interface.
- Responsive across devices.
- Scalable architecture.
- Reusable React components.

Limitations

- No real blockchain integration.
- No backend database.
- Wallet functionality is simulated.
- No real payment gateway.
- Limited multi-user support.

Future Enhancements

- Blockchain integration.
- Smart contract deployment.
- NFT minting support.
- User authentication.
- Cloud database integration.
- Real-time notifications.
- Payment gateway support.
- Mobile application development.

Conclusion

CopyrightFlow successfully demonstrates a modern digital asset licensing and royalty management platform using React and contemporary frontend technologies. The project provides a practical solution for asset management, royalty tracking, marketplace operations, and analytics visualization. Through its modular architecture and scalable design, the platform establishes a strong foundation for future blockchain and enterprise-level enhancements.

References

1. [React Documentation](#)
2. [Vite Documentation](#)
3. [Tailwind CSS Documentation](#)
4. [Framer Motion Documentation](#)
5. [Recharts Documentation](#)
6. [React Router Documentation](#)